A Spectral Hierarchical Machine Learning for Predicting Arsenic Concentration in Farmland Soil Using Sentinel-2 Imagery

Li Wang, Yong Zhou, Zehan Zhou, Shangrong Wu, *Senior Member, IEEE*, Lang Xia, Yan Zha, and Peng Yang

***Abstract*— Accurately predicting arsenic (As) concentration in farmland soil on a large scale is essential for effectively preventing and managing soil pollution in agricultural areas, thereby safeguarding food security. Multispectral imaging presents a cost-effective and efficient method for monitoring As concentration across extensive farmland regions. Nevertheless, the underlying process and mechanisms determining the relationship between As concentration in farmland soil and spectral data remain uncertain. The primary aim of this study was to evaluate whether employing a hierarchical strategy (based on soil organic matter (SOM) and pH) results in more accurate prediction of As concentration in farmland soil than those employing nonhierarchical (global) models. Our results show that with respect to global models, the best prediction of As concentration was achieved using the convolutional neural network (CNN) model (validated ratio of the model performance to the interquartile distance (RPIQ) = 2.50), followed by the Cubist model (validated RPIQ = 2.19) and the extreme learning machine (ELM) model (validated RPIQ = 2.15). After SOM-based hierarchization, the Cubist model exhibited the highest prediction accuracy (validated coefficient of determination (R^2) = 0.73), representing a 0.02 improvement in the R^2 compared with the that of global CNN model. The clay mineral ratio (CMR) was identified as the most important variable for predicting As concentration. Notably, the identification of high As concentration in the central old town areas underscores the importance of early soil contamination risk warnings on arable lands. These findings indicate that SOM-hierarchical machine learning models could serve as an effective approach to address the influence of soil environmental complications on spectral prediction of As concentration in farmland soil. By implementing this proposed method, soil environmental monitoring efforts can be significantly improved.**

***Index Terms*— Farmland soil, machine learning, multispectral image, soil arsenic, spectral hierarchization.**

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I. Introduction

As THE global population grows and agricultural production demands increase, the environmental quality of agricultural land is becoming vital for human health and sustainable development [1], [2]. Soil pollution, particularly in cultivated areas, poses a significant threat to food security and the ecological environment [3], [4]. Among various contaminants, heavy metal pollution is particularly concerning due to the nondegradable nature and high toxicity of these elements [5], [6]. Their accumulation in farmland soil not only impairs crop growth but also poses serious health risks through bioaccumulation in the food chain [7]. For instance, arsenic (As), a highly toxic heavy metal, is increasingly contaminating farmland due to agricultural practices, mining, industrial waste, and the use of fertilizers and pesticides [8]. In some agricultural-intensive regions, the As concentration in soil exceeds safe thresholds. Notably, approximately 2.7% of arable land in China is affected by As contamination [9]. Hence, efficient and accurate prediction and monitoring of As concentration in cropland soil are crucial for effective soil pollution control and agricultural environmental management.

Nevertheless, accurately predicting As concentration in farmland soil still presents several significant challenges. First, soil As distribution exhibits significant spatial heterogeneity, indicating that its concentration is affected by a range of factors beyond geographic location, such as soil type, topography, hydrological conditions, and human activities [10]. Although traditional field sampling and laboratory analysis can yield accurate point data, their high costs and time-consuming nature render large-scale monitoring difficult [11]. Recently, remote sensing technology has been extensively utilized for tracking soil heavy metal pollution. By capturing extensive spectral information, researchers can develop predictive models that link spectral characteristics to heavy metal concentrations [12], [13]. However, the complex nonlinear characteristics of soil

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spectral data, coupled with its sensitivity to noise, pose substantial challenges in constructing high-precision predictive models.

Remote sensing technology, especially multispectral imaging, provides a nondestructive, rapid, and wide-ranging solution for predicting and monitoring heavy metal contamination in farmland soil [14], [15]. By capturing the reflectance spectra of land features across multiple specific bands, multispectral imaging effectively detects potential heavy metal pollution in these soils [16]. Previous studies have shown that heavy metals such as As display significant spectral characteristics at certain wavelengths, which facilitates prediction analysis using remote sensing techniques [17], [18]. Moreover, satellite multispectral remote sensing technology facilitates more precise and timely identification and assessment of soil heavy metal contamination by efficiently covering extensive areas and offering real-time monitoring capabilities [19], [20].

Despite the convenience that remote sensing technology offers for monitoring heavy metal pollution, soil spectral data are often affected by noise and redundant information, making it difficult to achieve optimal predictive performance when directly applied to heavy metal prediction models [21], [22]. Spectral hierarchization has been introduced as a new preprocessing technique to manage the raw spectral data. This method divides the spectral data into different layers, with each layer corresponding to specific spectral features, thus effectively extracting key information related to heavy metal concentrations [23], [24]. This approach not only helps to reduce data redundancy and noise but also allows for capturing spectral variations associated with As concentration at different levels [25]. Consequently, it provides a more streamlined and effective input for subsequent soil heavy metal spectral modeling, potentially improving predictive accuracy. Moreover, variations in soil spectra can be reflected by the relationship between soil and the environment [26], [27]. Attempting to hierarchize soil heavy metals using environmental covariates is a promising strategy for predicting soil heavy metals. However, for agricultural land, the effectiveness of different hierarchization strategies in capturing the covariation between heavy metals and spectral data remains unclear. Identifying the most suitable hierarchization method may help reduce data variability and further enhance predictive performance.

In the realm of soil heavy metal prediction, machine learning techniques, such as the extreme learning machine (ELM), backward-propagation neural network (BPNN), and random forest (RF), have gained traction in recent studies [12], [19]. Although these models have enhanced prediction accuracy, they still encounter limitations in handling complex nonlinear features and high-dimensional data. In recent years, advanced deep learning methods, especially the convolutional neural network (CNN), have achieved substantial progress in processing and analyzing remote sensing data [28]. Leveraging its robust feature extraction capabilities and strength in managing nonlinear relationships, CNN can automatically learn and extract intricate information from spectral data, facilitating precise predictions even in complex environments [29]. When compared to traditional machine learning models, the CNN excels in progressively extracting local features from spectral data through layered convolution operations while maintaining spatial information [30]. Furthermore, by incorporating pooling layers and fully connected layers, the CNN can effectively minimize noise interference and enhance the model's generalization ability [31]. Consequently, the CNN exhibits significant potential for application in the field of remote sensing-based soil heavy metal prediction. Therefore, it is worth addressing whether the integration of the above spectral hierarchical strategy with machine learning algorithms can result in improved spectral predictions of soil heavy metals.

A significant knowledge gap exists in understanding how to effectively integrate different spectral hierarchization approaches with machine learning models to extract complex spectral features associated with As concentration in farmland soil. To address this gap, this study proposes an innovative framework that combines spectral hierarchization with machine learning models using satellite remote sensing imagery to enhance the accuracy of predicting As concentration in farmland soil. The primary contributions of this study are presented as follows.

1) The spectral hierarchization approach effectively eliminates redundant information and enhances data quality through hierarchical segmentation and processing of the original spectral data. In addition, this approach highlights spectral features related to As concentration, making them more effective in subsequent modeling and improving the accuracy of spectral predictions.
2) By employing multilevel feature extraction, machine learning models can effectively capture the intricate nonlinear relationships between As concentration and various soil variables. Furthermore, the preprocessing step of spectral hierarchization minimizes noise interference, and when combined with the model's inherent robustness, it significantly enhances the reliability of the predictive outcomes.
3) The spectral hierarchization machine learning model has exhibited remarkable advantages in mapping soil heavy metals using satellite remote sensing imagery. Research conducted in key grain-producing regions of China shows that this model provides cutting-edge performance for this application.

II. Materials and Methods

A. Study Area

Xiangzhou District is located within Xiangyang City, Hubei Province, in central China, positioned along the central stretch of the Han River (111°44′–112°23′ E and 31°46′–32°28′ N) [see Fig. 1(a) and (b)]. The region encompasses a total land area of 2306 km^2, with the cultivated area reaching 1700.53 km^2 in 2019, representing approximately 73.74% of the total land area. It features diverse topography, characterized by low mountain hills in the north, while the remaining areas are predominantly shaped by the alluvial plain of the Han River [see Fig. 1(c)]. According to Chinese Soil Taxonomy [32], the study area encompasses six different soil categories. Yellow-cinnamon soils, submergenic paddy soils,

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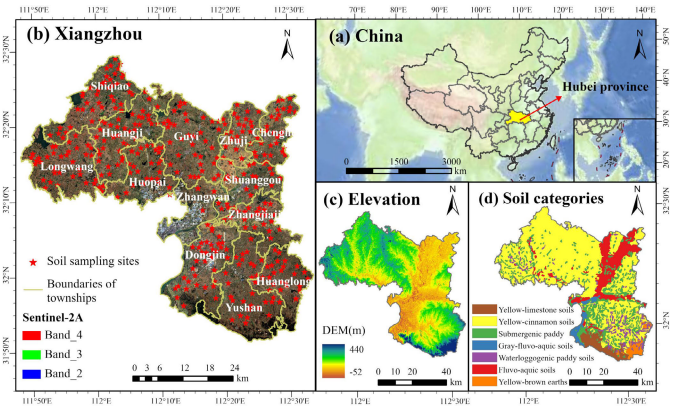


Fig. 1. Geographic location and soil sampling sites of the study area (b) in China (a), its elevation (c), and soil categories (d).

and fluvo-aquic soils are the main soil orders [see Fig. 1(d)]. The region features a subtropical humid monsoon continental climate, characterized by an annual temperature ranging from 15.3 °C to 15.8 °C and a mean precipitation of 878.3 mm. Good climatic conditions provide suitable sites for agricultural production. The major crops cultivated in the region include rice, wheat, and maize.

Xiangzhou District, a key transportation hub in Hubei Province, boasts significant economic development, ranking among the province's leaders. This strategic location, at the convergence of major highways (Fuyin, Erguang) and railways (Xiangyu, Jiaoliu, Menghua), as well as high-speed rail lines connecting Han-Shenzhen and Zheng-Yu, supports a diverse manufacturing sector. Three key industries—automotive and parts, agricultural product processing, and equipment manufacturing—each generate annual production exceeding 10 billion Chinese Yuan. However, an environmental investigation conducted in 2019 by the environmental protection agency identified 20 potential pollution enterprises within the district, including Camel Group Xiangyang Battery Company Ltd., Xiangyang Te Yi Insulation Materials Engineering Company Ltd., and Dai Ruimik Xiangyang Battery Separator Company Ltd. The presence of transportation networks and potential polluting industries may lead to varying degrees of contamination in nearby water bodies and soils, ultimately contributing to heavy metal pollution in the soil of surrounding farmlands.

B. Datasets

1) Soil Sampling and Laboratory Analysis: To comply with the Technical Specifications for Monitoring the Environmental Quality of Farmland Soil (NY/T 395-2012) [33], we implemented a grid-based sampling strategy to ensure comprehensive and representative coverage of the study area. This approach incorporated crucial environmental factors, including topography, cropland distribution, distance from industrial areas, and road proximity, to optimize sampling point placement. The study area was divided into 2 × 2 km grids, with individual cropland plots within each grid serving as sampling units. To enhance sampling representativeness in areas potentially impacted by localized pollution sources, we increased sampling density around key soil pollution monitoring enterprises. A total of 373 soil samples were obtained from harvested cropland in October 2019 from bare plots at a depth of 0–20 cm with no crop residue on the soil surface to avoid mixed pixels [see Fig. 1(b)]. We used shovels to collect five subsamples within a 10 m radius for each sample and cleaned each subsample with ethanol to avoid cross-contamination. During sampling, handheld GPS devices were utilized to record the latitude and longitude coordinates of each sampling location. The collected soil samples were immediately placed in polypropylene-sealed bags to ensure their integrity and then quickly conveyed to the lab. The soil samples were dried in the open air to remove moisture content before being processed through a sieve to ensure uniformity in texture for subsequent analyses.

The soil As concentration was assayed via inductively coupled plasma-atomic emission spectroscopy [34]. Approximately 0.1 g of sample (weighed to the nearest 0.0001 g) was digested in a polytetrafluoroethylene (PTFE) beaker using a mixture of hydrofluoric acid (HF) and perchloric acid ($HCLO_4$). The solution was heated gently until white fumes evolved, ensuring complete sample decomposition. After cooling, nitric acid (HNO_3) was added, followed by thorough water rinsing. The digested solution was then quantitatively transferred to a 100-mL volumetric flask, diluted to the mark, and mixed well. The As concentration was subsequently measured using an iCAP 6300 ICP-AES (Thermo Fisher Scientific, USA) under optimized conditions. A simultaneous blank measurement was performed for background correction.

The soil organic matter (SOM) content was determined using the dichromate oxidation method [35]. Briefly, an excess of a standard potassium dichromate ($K_2Cr_2O_7$) solution in concentrated sulfuric acid (H_2SO_4) was added to the soil sample. The mixture was heated to oxidize the organic carbon. The remaining unreacted dichromate was then titrated with a standard ferrous sulfate ($FeSO_4$) solution. The amount of $K_2Cr_2O_7$ consumed during the oxidation process was used to estimate the SOM content of the soil sample.

The soil pH was determined using the potentiometric technique [36]. Prior to measurement, the glass and glycerol electrodes were calibrated with standard buffer solutions of pH 6.88 (phosphate) and 9.22 (borate). Subsequently, the electrodes were immersed in a soil suspension prepared with a water-to-soil ratio of 2.5:1. The pH was recorded after the electrode readings stabilized.

2) Remote Sensing Image Acquisition and Processing: In this study, Sentinel-2A imagery was used to provide spectral information. Cloud-free Sentinel-2A images collected on October 26, 2019, were obtained from European Space Agency's Sentinels Scientific Data Hub (https://scihub.copernicus.eu/). The atmospheric calibration of the Sentinel-2A images was performed using the Sen2Cor plug-in, which played a crucial role in precisely obtaining the surface reflectance values. The images were subsequently cropped based on vector boundaries of the study area, enabling a focused analysis of the specific region of interest and removing irrelevant background information. We utilized the coordinates from the surveyed ground control points on the cropped images to perform geometric correction, ensuring

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precise alignment with actual locations. We modified the resolution of the Sentinel-2A images to 10 m by employing the nearest neighbor interpolation method, enhancing the detail and clarity of the resulting images.

The corresponding raw reflectance spectrum was retrieved from the latitude and longitude of the soil sample points. Recognizing the weak spectral sensitivity of soil As, spectral transformations were systematically performed to effectively reduce noise and enhance differences in spectral features observed between the various samples. Consequently, a first-order differential (FD) transformation was imposed on the raw spectral reflectance, thereby improving the overall clarity and interpretability of the spectral data. This transformation not only allows for a more accurate assessment of the soil's chemical properties but also facilitates reliable comparisons across different samples, ultimately contributing to a deeper understanding of spatial contamination patterns [15].

C. Features Extraction

1) Spectral Features: A diverse set of features was selected to predict soil As concentration through pixel extraction and band calculations. This selection included spectral values as well as various spectral indices, such as vegetation indices and soil quality indices (see Table I). Spectral values encapsulate information from different bands, providing insights into land surface characteristics and effectively predicting soil As concentration by capturing subtle variations in reflectance [12], [13]. Vegetation indices were employed to evaluate plant health and growth under heavy metal stress, effectively revealing physiological responses to toxins and enabling the monitoring of contamination impacts on both plant vitality and overall ecosystem health [37]. Soil quality indices significantly influence soil heavy metal dynamics by enhancing metal retention and reducing mobility, thus mitigating their toxicity and environmental impact [38]. The spectral values were obtained, and the spectral indices were computed using ArcGIS 10.7 (Environmental System Research Institute, Redlands, CA, USA).

2) Feature Selection: Feature selection plays a crucial role in estimating soil properties by eliminating unnecessary or redundant data [46]. This allows the model to concentrate on the most relevant predictors, leading to simpler models that are more manageable and easier to analyze. In addition, reducing complexity aids in mitigating overfitting, a frequent issue in high-dimensional datasets. Pearson correlation analysis is a statistical method that assesses linear relationships between variables, aiding in the identification of important features and the elimination of redundancies to improve model performance and simplify complexity [47]. In this study, sensitivity bands and spectral indices were determined by the Pearson correlation analysis at a significance level of 0.05 between them and the soil As concentration. Then, the spectral transformed data and spectral indices with the highest correlations were chosen as the predictor variables for modeling.

D. Modeling Strategy

1) Training and Validation Datasets: We grouped the 373 samples into two groups via a concentration ordering method to support model training and evaluation. A training set, consisting of 75% of the samples (280 samples), was used to develop and calibrate the model, while a validation set, representing the remaining 25% (93 samples), assessed the model's predictive accuracy and generalizability. This approach was implemented to ensure that the As concentration range of the training dataset exceeds that of the validation dataset, resulting in approximately analogous data distributions. The selection procedure for As dataset was as follows: 373 samples were ranked by ascending As concentration and then organized into 93 strata. One sample in each stratum was designated the validation dataset, and the remaining samples were designated the training dataset. Levene's test

TABLE I
EXTRACTED SPECTRAL FEATURES

Type	Feature	Formula	Reference
Spectral values	Bands (B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12)	-	-
Vegetation indices	NDVI	$\frac{B8-B4}{B8+B4}$	Rouse et al. [39]
	EVI	$\frac{2.5(B8-B4)}{B8+6B4-7.5B2+1}$	Huete et al. [40]
	SAVI	$\frac{1.5(\mathrm{B8-B4})}{\mathrm{B8+B4+0.5}}$	HUETE [41]
	NDVIre1	$\frac{B8-B5}{B8+B5}$	Gitelson and Merzlyak [42]
	NDVIre2	$\frac{B8-B6}{B8+B6}$	
	NDVIre3	$\frac{B8-B7}{B8+B7}$	
	CIre	$\frac{B7}{B5}-1$	Gitelson et al. [43]
Soil quality indices	CMR	$\frac{B11}{B12}$	Michalski et al. [44]
	IOR	$\frac{B4}{B2}$	Liu and Mason [45]

Note: B2, B3, and B4 represent Blue, Green, and Red bands, respectively. B5, B6, and B7 belong to the Vegetation Red Edge band. B8 and B8A belong to the Near-infrared band, while B11 and B12 belong to the Shortwave Infrared band. NDVI stands for Normalized Difference Vegetation Index, EVI for Enhanced Vegetation Index, SAVI for Soil-Adjusted Vegetation Index, NDVIre1, NDVIre2, and NDVIre3 for Normalized Difference Vegetation Index Red Edge 1, 2, and 3, respectively. CIre represents Chlorophyll Indices Red Edge, CMR denotes Clay Minerals Ratio, and IOR signifies Iron Oxide Ratio.

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was computed in R 3.8 software (University of Auckland, Auckland, New Zealand) at a significance level of 0.05, and the car package was used to examine the homogeneity of variance between the training and validation datasets.

2) Global Modeling: Six regression methods [e.g., ELM, RF, light gradient boosting machine (LightGBM), Cubist, extreme gradient boosting (XGBoost), and CNN] were established, and their As prediction performance was assessed and contrasted.

The RF (randomForest package in R) is a decision tree-based machine learning model that has been utilized for its advantages in handling multivariate nonlinear data to predict soil heavy metals [48]. The RF model has no restrictions associated with the input data, is highly tolerant of outliers, and is not prone to overfitting [49]. The ELM (elmNNRcpp package in R) is used to train single-hidden-layer feedforward neural networks because of its fast training speed and resistance to becoming trapped in local optima [50]. This method randomly assigns input layer weights and node biases during the training process to obtain distinct optimal solutions [51]. The LightGBM (lightgbm package in R) is a gradient boosting framework utilizing tree-based learning models and a leafwise tree growth strategy, leading to faster training times and lower memory usage compared to those of traditional depth-wise tree growth strategies [52]. The XGBoost (xgboost package in R) is an integrated learning method optimized based on gradient boosting decision trees. This method is able to computations via parallel learning and improve the prediction accuracy by improving the objective function [53]. The Cubist (Cubist package in R) is a powerful rule-based decision tree method for the categorical regression of continuous variables. This method automatically partitions the input space and provides a good solution to the high-dimensional data problem [54]. To achieve the optimal model performance, we implemented a comprehensive hyperparameter tuning strategy. This process utilized fivefold cross-validation repeated ten times in combination with a grid search approach. The key machine learning models were systematically fine-tuned by optimizing their critical hyperparameters, including RF (ntree, mtry, and nodesize), ELM (nchild and actfun), LightGBM (learning_rate, num_iterations, num_leaves, and feature_fraction), XGBoost (eta, max_depth, subsample, and colsample_bytree), and Cubist (committees and neighbors). Detailed descriptions of the hyperparameters and their corresponding optimization ranges are provided in Table II.

The CNN is an important technique for deep learning with the advantages of weight sharing and automatic feature extraction [55]. This method is a convolutional computational and deeply structured feedforward neural network that comprises an input layer, a convolutional layer, an activation layer, a pooling layer, and a fully connected layer [29]. The convolutional layer is the core component of the CNN. In this layer, the implicit features of the data are automatically extracted through local sensing and a parameter sharing mechanism. Pooling is a nonlinear dimensionality reduction method that involves secondary feature extraction for the feature values output from the convolutional layers and suppresses network overfitting. In this study, a 1-D CNN model was constructed

TABLE II
DETAILED DESCRIPTIONS OF THE HYPERPARAMETERS AND THEIR CORRESPONDING OPTIMIZATION RANGES

Models	Hyperparameters	Search range
RF	ntree	[100, 1000]
	mtry	[1, 8]
	nodesize	[5, 10]
ELM	nchild	[10, 50]
	actfun	['sig', 'sin', 'relu', 'purelin']
LightGBM	learning_rate	[0, 1]
	num_iterations	[100, 500]
	num_leaves	[10, 50]
	feature_fraction	[0, 1]
XGBoost	eta	[0.01, 0.3]
	max_depth	[1, 10]
	subsample	[0.5, 1]
	colsample_bytree	[0.5,1]
Cubist	committees	[1, 100]
	neighbors	[0, 9]

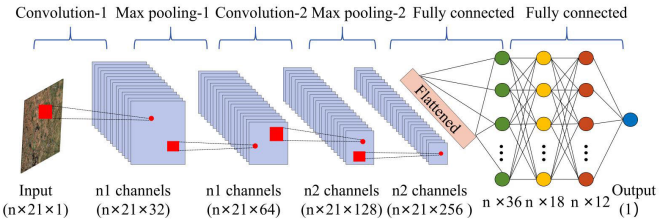


Fig. 2. Architecture of the CNN model.

(see Fig. 2). This model has three convolutional pooling layers. The convolutional kernel size was assumed to be 5, with a step size of 1. The pooling layers had 8, 16, and 32 convolutional kernels. A rectified linear unit (ReLU) activation function was employed in every convolutional layer. There were two hidden layers in the fully connected layer, and the number of neurons in the first hidden layer and second hidden layer was 32 and 1, respectively. The CNN was optimized using the adaptive moment estimation (Adam) algorithm as the loss function. The initial learning rate was assumed to be 0.05, with an epoch of 150 and a batch size of 64. We employed Shapley additive explanation (SHAP) to reveal the relative importance of each input variable. The CNN was built based on the TensorFlow platform and implemented in Python 3.11 (Centrum Wiskunde & Informatica, Amsterdam, The Netherlands). We acquired SHAP values in Python 3.11 by utilizing the SHAP library.

3) Spectral Hierarchical Strategies: Two different hierarchical strategies (i.e., SOM classification and pH classification) were used. The As concentration in farmland soil is generally considered low, and the SOM exhibits a certain adsorption effect on the accumulation of As [56]. Consequently, the training dataset was divided into three categories according to the SOM content: SOM $<$ 10 g/kg, 10 g/kg $-$ SOM $-$ 20 g/kg, and SOM $>$ 20 g/kg. The pH influences the spatial distribution of the As concentration by altering the morphology of As and

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its transport capacity [57]. Accordingly, the training dataset was divided into three categories based on the pH value: pH < 5.5, 5.5- pH- 6.5, and pH > 6.5. Soil samples from the validation dataset were assigned to groups identified through SOM and pH-based hierarchical clustering. Subsequently, soil As prediction models within each of these groups were developed using the abovementioned six regression methods.

E. Prediction Performance and Mapping

The coefficient of determination (R^2), the root-mean-square error (RMSE), and the ratio of model performance to the interquartile distance (RPIQ) were employed to evaluate the prediction performance. R^2 measures the proportion of variance in the dependent variable that can be explained by the independent variables. RMSE quantifies the average deviation between predicted and observed values. RPIQ evaluates the model's robustness and ability to generalize across different data subsets [58]. Higher R^2 and RPIQ values, alongside lower RMSE values, demonstrate superior predictive performance of the model, reflecting its capacity to explain a significant portion of the variance in the observed data. In addition, the 1:1 line, which includes Lin's concordance correlation coefficient (LCCC), was used to measure the extent to which the measured As concentration deviates from the predicted value [59]. The formulas are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{1}$$

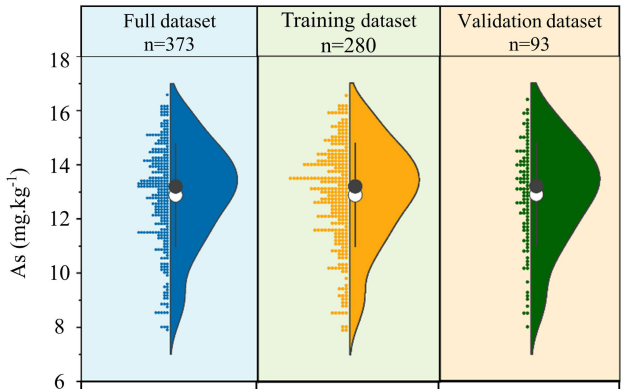
$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}} \tag{2}$$

$$\text{RPIQ} = \frac{Q_3 - Q_1}{\text{RMSE}} \tag{3}$$

$$\text{LCCC} = \frac{2\rho\sigma_y\sigma_{\hat{y}}}{(\bar{y} - \bar{\hat{y}})^2 + \sigma_y^2 + \sigma_{\hat{y}}^2} \tag{4}$$

where n is the number of samples; y_i and $\hat{y}_i$ are the measured and predicted values, respectively; $\bar{y}$ is the mean of the measured values; Q_1 and Q_3 represent the 25th and 75th percentiles, respectively; ρ denotes the Pearson correlation coefficient between measured and predicted values; σ_y and $\sigma_{\hat{y}}$ represent the variances of measured and predicted values, respectively; and $\bar{\hat{y}}$ is the mean of the predicted values.

The optimal trained model, developed using the R 3.8 platform, was employed to estimate the As concentration in cropland soil across a defined grid in the study area, ensuring that the predictions were both accurate and reliable. Following this, the Bayesian Kriging (BK) method was utilized to assess the predictions between every grid point, a technique that effectively accounts for spatial correlation and uncertainty in the data. As a result, this approach enabled the generation of a comprehensive map that illustrates the spatial distribution of estimated soil As concentration throughout the study area. Furthermore, by integrating land use change survey data, we extracted farmland plots from the study area and applied them as a mask to the overall soil As concentration map, thereby obtaining the spatial distribution of As concentration specifically within farmland areas throughout the study area.



	Full dataset n=373	Training dataset n=280	Validation dataset n=93
Range	7.91-16.60	7.91-16.60	8.01-16.4
Mean	12.88	12.878	12.886
Std.Dev	1.904	1.904	1.913
CV(%)	14.78	14.78	14.85
Skewness	-0.51	-0.51	-0.52
Kurtosis	-0.21	-0.20	-0.20
p-value	Anderson-Darling test 0.10	t-test 0.97	Levene's test 0.97	Bonett's test 0.95
Soil standard[a]	30(pH≤5.5)	30(5.5 < pH≤6.5)	25(6.5 < pH≤7.5)	20(pH > 7.5)

Fig. 3. Summary statistics for measured soil As. Note: the white and black dots indicate the mean and the median values, respectively. Std.Dev stands for standard deviation. CV stands for coefficient of variation. [a]Soil environmental quality risk control standard for soil contamination of agricultural land (GB15618-2018).

The resulting refined map of As contamination in farmland soil is crucial for identifying high-risk areas, informing best agricultural practices, and guiding policy decisions related to soil health and environmental safety.

III. Results

A. Exploratory Analysis and Soil Spectral Characteristics

The As concentration across the dataset exhibited significant variability extending from 7.91 to 16.60 mg/kg, with a mean and a standard deviation (SD) of 12.88 and 1.904 mg/kg, respectively (see Fig. 3). Similar mean and SD values for As were observed in the validation dataset compared to the training dataset. In addition, the large coefficient of variation (CV) for As indicated substantial data heterogeneity. The soil pH ranged from 4.20 to 8.18, with 98% of the samples having a pH lower than 7.5. Although the mean As values did not exceed the risk threshold values, suggesting generally safe levels of As pollution in farmland soil (see Fig. 3), the mean As concentration slightly exceeded the background value [60]. The Anderson–Darling (AD) test, along with skewness and kurtosis values, indicated a normal distribution of As concentration. Furthermore, the t-test, Levene's test, and Bonett's test confirmed reasonable division of the dataset ($p > 0.05$).

With respect to the raw spectra, the spectral absorptive signature of the Sentinel-2A data was augmented following FD treatment [see Fig. 4(a) and 4(b)]. Across the entire spectral range, the spectral correlation between the raw image and the FD data ranged from −0.46 to −0.17 and from −0.49 to 0.30, respectively [see Fig. 4(c) and (d)].

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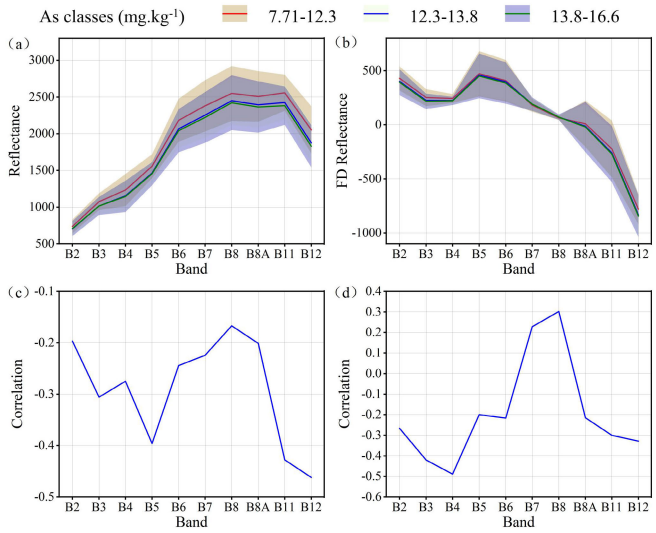


Fig. 4. (a) Raw spectra and (b) first-derivative spectra under three As classes. Spectral correlations for (c) raw image data and (d) first-derivative data.

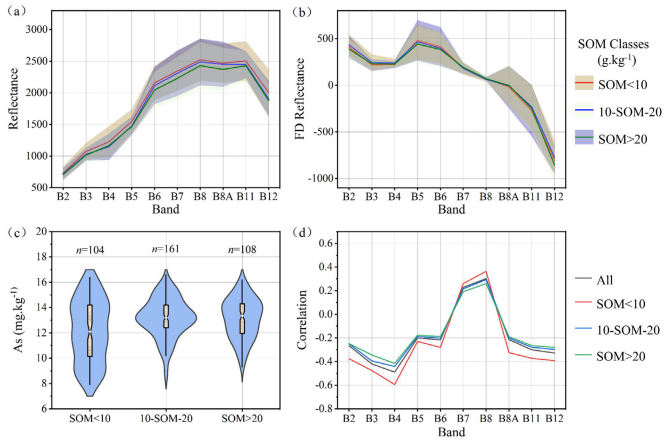


Fig. 5. Hierarchization of spectral data with SOM: (a) raw reflectance, (b) FD reflectance, (c) violin plots of As with SOM classes, and (d) correlation among As and reflection spectra with various SOM classes.

B. Spectral Hierarchization and Data Signatures

Hierarchization of all sample spectral data using the SOM revealed a consistent decreasing trend in the reflection spectra as SOM content increased [see Fig. 5(a)], whereas comparable mean values of As associated with the three SOM categories were observed [see Fig. 5(c)]. In the case of the FD spectra of As, higher absorbance profiles typically correlated with higher SOM content [see Fig. 5(b)]. Concerning the spectral correlations [see Fig. 5(d)], the absolute correlation values for SOM values less than 10 g/kg were consistently greater than those for all samples, while the correlation values for SOM $>$ 20 g/kg and 10 g/kg $-$ SOM $-$ 20 g/kg decreased after SOM-based hierarchization.

The reflectance spectra of various pH categories exhibited similar spectral curves but varied in spectral intensities [see Fig. 6(a) and (b)]. Apart from the distinct absorption characteristics at B12 (a center wavelength of 2202 nm), two additional absorption valleys were identified at B4 (a center wavelength of 665 nm) and B8A (a center wavelength of

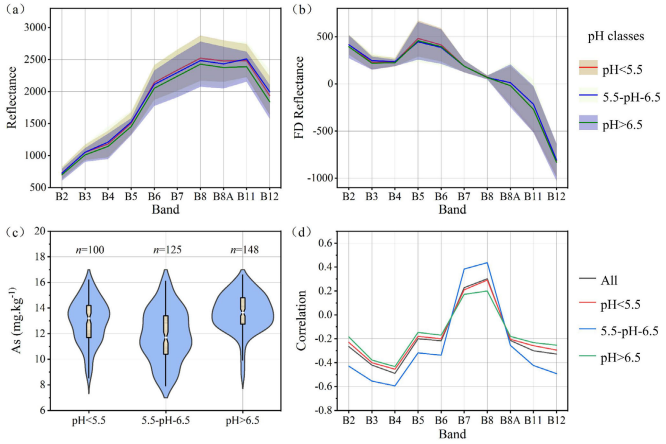


Fig. 6. Hierarchization of spectral data with pH: (a) raw reflectance, (b) FD reflectance, (c) violin plots of As with pH classes, and (d) correlation among As and reflection spectra with various pH classes.

865 nm), indicating the presence of goethite and haematite. The mean As values across various pH categories were overall comparable [see Fig. 6(c)]. Following pH-based hierarchization, the spectral correlations (absolute values ranging from 0.25 to 0.59) of samples with pH values between 5.5 and 6.5 ($n = 125$) were found to be greater than those of all samples ($n = 373$) across the whole spectral region [see Fig. 6(d)].

C. Selection of the Optimal Spectral Variable

A significant negative correlation was observed between the As concentration and raw reflectance [see Fig. 4(c)]. Subsequent FD transformation of the raw reflectance data notably improved correlation in certain bands [see Fig. 4(d)], indicating that FD can effectively accentuate the soil As reflectance characteristics. Bands with higher FD reflectance correlations were predominantly concentrated at B4 (a center wavelength of 665 nm), B8 (a center wavelength of 835 nm), B8A (a center wavelength of 865 nm), and B12 (a center wavelength of 2202 nm) (see Table III), likely attributed to specific elemental oscillations [61]. In addition, spectral indices correlations were consistently greater than those of FD reflectances (see Table III). The clay mineral ratio (CMR) exhibited the highest correlation among all spectral variables, with a calculated value of 0.55 (see Table III). The spectral variables listed in Table III were utilized as input values for the model, indicating their importance in predicting soil As concentration.

D. Spectroscopic Modeling

Regarding the global training, the CNN model (validation $R^2 = 0.71$) achieved the highest validation prediction performance (see Table IV). It was closely followed by the Cubist model (validation $R^2 = 0.65$), while the LightGBM model yielded the lowest result (validation $R^2 = 0.59$). Notably, the ELM and RF models exhibited commendable performance, with validated RPIQ values of 2.15 and 2.14, respectively. However, all six models, especially the XGBoost and LightGBM models, exhibited underestimation in predicting samples with high As concentration ($>$14 mg/kg) (see Fig. 7).

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TABLE III
HIGHEST SPECTRAL VARIABLE CORRELATED WITH AS CONCENTRATION

FD reflectance	|R|	Spectral indices	|R|
B8A	0.49**	CMR	0.55**
B12	0.42**	IOR	0.45**
B8	0.33**	NDVIRE2	0.35**
B4	0.32**	EVI	0.31**

Note: ** Significant at $p < 0.01$.

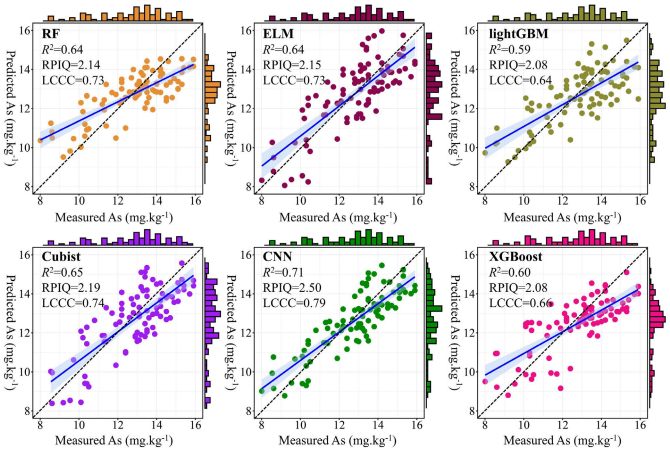


Fig. 7. Scatter plots of the measured versus the predicted As obtained from the six global models. The top and right histograms represent measured As and predicted As, respectively. The black dashed lines denote 1:1 lines, the blue solid lines denote the regression lines, and the blue areas denote the 95% prediction band.

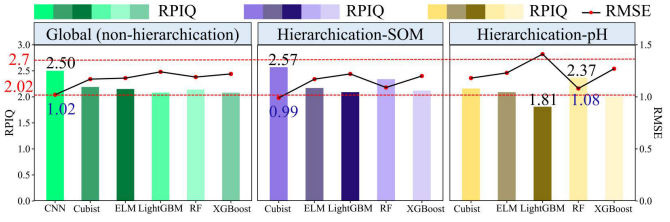


Fig. 8. RPIQ and RMSE for validation datasets relying on diverse models.

Due to the limited number of samples available after spectral hierarchization in certain categories, the CNN model was not employed (see Table IV). The remaining models underwent separate calibration and validation using samples from each category. Subsequently, the modeling outputs from each validation were merged to evaluate the overall predictive performance. Among the SOM-hierarchized models, the Cubist model exhibited the highest predictive performance (validation RPIQ = 2.57 and validation RMSE = 0.99), marginally surpassing the top-performing global CNN model (see Fig. 8).

Comparatively, the predictive results of the pH-hierarchized model (validation R^2 = 0.59–0.70) generally lagged behind those of the SOM-hierarchized model (see Fig. 9). In addition, the best validation accuracy achieved by RF within the pH-hierarchical model (validation R^2 = 0.70) was also lower than that of the global CNN model (validation R^2 = 0.71) (see Fig. 9).

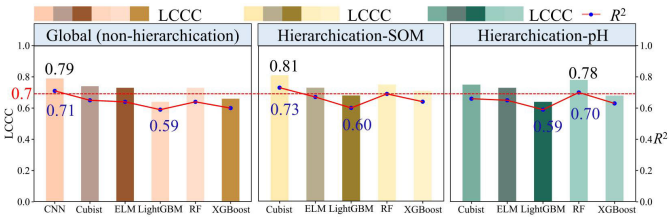


Fig. 9. R^2 and LCCC for validation datasets relying on diverse models.

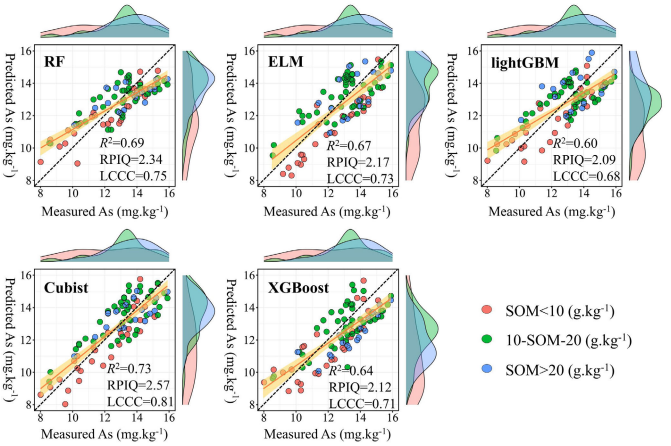


Fig. 10. Scatter plots of the measured versus the predicted As obtained from the SOM-hierarchized models. The upper and the right edge density plots represent measured and predicted As, respectively. The black dashed lines denote 1:1 lines, the red solid lines denote the regression lines, and the yellow areas denote the 95% prediction band.

To visualize the predictive performance of the independent validation dataset, the validation results for each model within each hierarchized strategy were obtained and are displayed in Figs. 10 and 11. In comparison to those of the other four models, the scatter distributions of the As concentration measured and predicted by the SOM-hierarchized Cubist model were closer to the 1:1 line (see Fig. 10). Within the pH-hierarchized strategy, the RF model demonstrated the best performance, with its regression line exhibiting the closest proximity to the 1:1 line (see Fig. 11) and the highest LCCC among the five models (see Fig. 9).

In addition, we elucidated the significance of each spectral variable for predicting As using the global CNN model, representing a typical black-box model. The CNN model identified the CMR as the most crucial spectral variable for predicting As, followed by B8A [see Fig. 12(a) and (b)]. The importance of iron oxide ratio (IOR) ranked third, closely approaching that of B12.

E. Mapping and Spatial Distribution

After applying the BK interpolation, a distinct spatial resemblance was noted between the laboratory-measured concentration of As in farmland soil and those predicted using the SOM-hierarchized Cubist model for spectral prediction (see Fig. 13). Areas exhibiting high As concentration were predominantly clustered within the central old town regions of Dongjin, Yushan, and Zhangwan, whereas lower As concentration was primarily dispersed across the northern

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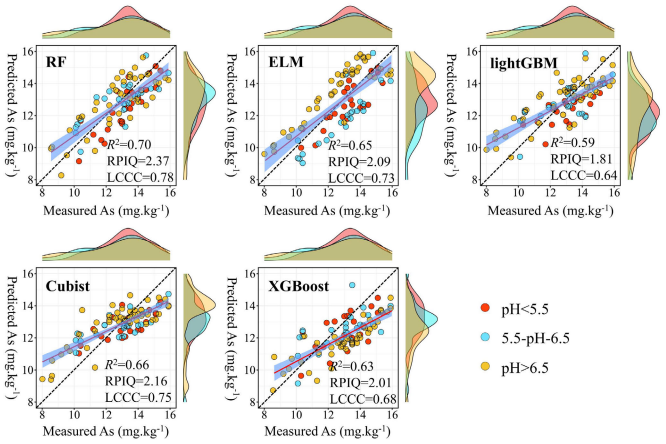


Fig. 11. Scatter plots of the measured versus the predicted As obtained from the pH-hierarchized models. The upper and the right edge density plots represent measured and predicted As, respectively. The black dashed lines denote 1:1 lines, the red solid lines denote the regression lines, and the blue areas denote the 95% prediction band.

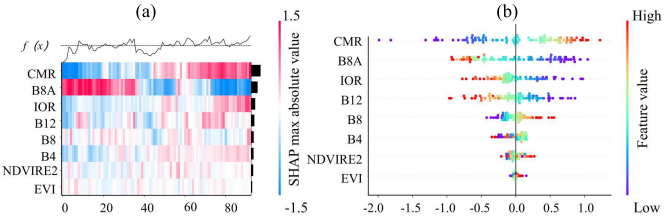


Fig. 12. (a) Heat map and (b) beeswarm plot of the relative importance of spectral variables extracted from the CNN model. The maximum absolute value of the Shapley additive explanation (SHAP) of the CNN model is computed with respect to all validation samples.

region. However, in contrast to the laboratory-measured As concentration, the SOM-hierarchized Cubist model tended to overestimate the low As concentration in the northern region (see Fig. 13). The semivariogram results, encompassing fit models and conversions, for both measured As and spectral-predicted As displayed a high degree of similarity (see Fig. 13), suggesting that the proposed optimal modeling method effectively captures the spatial variation in As within the study area.

IV. Discussion

A. Variability of Soil As and Its Sources

Heavy metals in unpolluted natural soil are usually present in minute amounts, resulting in low soil toxicity [62]. However, human activities can contribute to the accumulation of these elements, particularly in agricultural regions [15]. In the present study, the As concentration in farmland soil exhibited moderate variability (CV = 14.78%, for the whole dataset) and significant enrichment compared to background levels (see Fig. 3). This may be associated with different land use practices and the long-term application of chemical fertilizers and pesticides, many of which contain trace amounts of heavy metals such as As. Over time, these elements can accumulate in farmland soil, leading to elevated concentration and potentially increasing the risk of contamination [63].

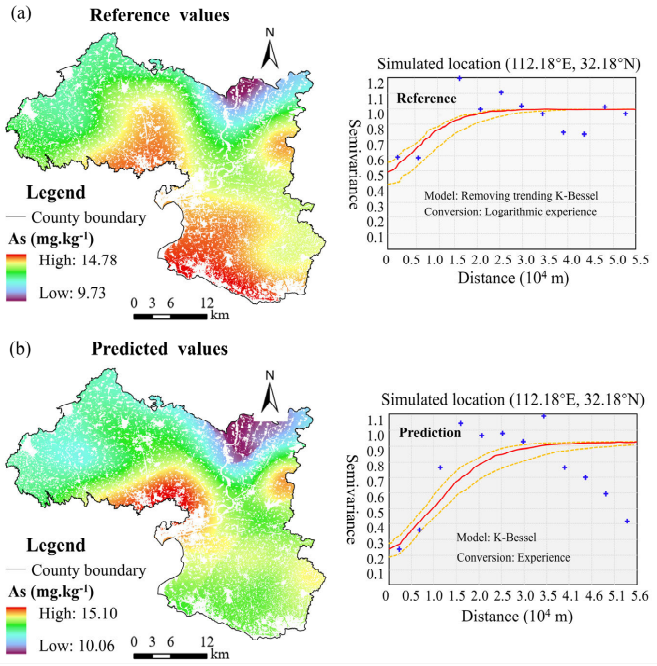


Fig. 13. Maps of As concentration in farmland soil throughout the study area: (a) derived from laboratory measurements and (b) predicted by the SOM-hierarchized Cubist model, with accompanying empirical variability plots.

In addition to agricultural practices, industrial activities, such as automobile manufacturing, iron and steel production, and road dust generation, are also potential sources of elevated As concentration. Atmospheric deposition of airborne particles from industrial sources can significantly contribute to farmland soil contamination. This is particularly evident in areas near industrial zones or highways, where airborne particles readily settle onto the land [64]. The combined influence of agricultural and industrial sources highlights the complex nature of As distribution in farmland soil. This complexity underscores the critical need for ongoing monitoring and management strategies to mitigate potential hazards to both public health and the environment.

B. Effects of Feature Contributions

The study underscores the crucial role of specific spectral values and spectral indices in accurately predicting soil As concentrations. Notably, both the maximum absolute value of SHAP values and the identification of optimal spectral variables revealed CMR, B8A, and IOR as key indicators of the soil's mineralogical and chemical composition, directly influencing As adsorption, mobility, and overall fate (see Table III and Fig. 12).

The CMR, the ratio of shortwave infrared 1 and shortwave infrared 2 bands, effectively captures radiation absorption in the 2.0–2.3-μm range, a spectral region highly sensitive to water-bearing minerals such as clay and alunite [65]. Clay minerals, abundant in many soils, possess high surface areas and inherent negative charges, exhibiting a strong affinity for As through electrostatic attraction and ligand exchange [66]. Alunite, a sulfate mineral, can also contribute to As sorption, particularly in acidic soil environments [67].

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The B8A, centered at 865 nm in the near-infrared region, is highly sensitive to the presence and abundance of organic matter within the soil [68]. Organic matter plays a pivotal role in As immobilization through complexation, forming stable complexes with As and reducing its bioavailability and mobility [69]. In addition, this band is associated with kaolin, a common clay mineral, and hydroxyl groups on mineral surfaces, both of which can contribute to As adsorption.

The IOR, the ratio of red to blue wavelengths, is particularly sensitive to the presence and abundance of iron oxides, such as goethite and hematite. These iron oxides exhibit a strong affinity for As, forming stable inner sphere complexes [70]. The oxidation state of iron within these minerals significantly influences As speciation and mobility. Moreover, IOR can serve as a visual indicator of iron oxide content, with areas of high iron oxide concentration often appearing brightly colored due to their strong red reflectance [71].

The interaction between soil As and these spectral variables is complex, involving a combination of adsorption mechanisms, including electrostatic attraction, ligand exchange, and specific surface complexation [72], [73]. Under specific geochemical conditions, As can form insoluble precipitates with iron oxides, sulfides, or other minerals, effectively immobilizing As within the soil matrix [66], [70]. Organic matter plays a crucial role in As immobilization through complexation, with the type and abundance of organic matter significantly influencing the strength and stability of these complexes [69]. The oxidation–reduction state of the soil environment, which can be indirectly inferred from spectral data, significantly impacts As speciation [74].

Overall, the fundamental absorption at important wavelengths is in good agreement with the band-specific assignments of As in soil, indicating that SHAP is appropriate for explaining the complicated CNN model. Our results show that SHAP is an effective technical tool for soil spectral analysis because it can explain complicated models and further enhance our understanding of spectral data.

C. Potential of Spectral Hierarchical Machine Learning for Predicting As

The CNN model achieved the highest prediction accuracy in the global models, even exceeding that of the majority of the models built by the spectral hierarchization (see Table IV), a finding consistent with that reported in other studies [75], [76]. Traditional machine learning methods (e.g., ELM, LightGBM, RF, and XGBoost) require spectral preprocessing, while the deep learning CNN method can automatically extract local characteristics of raw spectra [77], [78]. Chen et al. [79] suggested that the kernel function in the CNN model is similar to that used in the preprocessing of spectral data, offering the potential of building more generalized spectral prediction models. Unlike traditional modeling approaches that utilize the entire training dataset, the Cubist model generates a series of segmented linear models to address the nonlinear and potentially complex relationships between As concentrations and spectral data [80].

Existing studies have shown that the correlation between As and spectral information exhibits spatial dependence and site-specific heterogeneity over large areas, which could negatively impact the spectral prediction of As [21], [72]. Previous studies have suggested that the prediction of soil As based on large-scale soil spectra may result in decreased model accuracy compared to that from local-scale soil spectra [17], [73]. Soil spectral data encompass diverse information, including soil color, mineral composition, organic matter content, elemental content, soil moisture, and soil texture. Moreover, soil pH can influence the solubility, adsorption, and bioavailability of soil As through complex physical, chemical, and biological processes [81]. Although the average soil As concentration did not exhibit significant differences across the three pH levels, the hierarchization according to pH influenced the spectral correlation of soil As (see Fig. 6). Compared to those of the global models (i.e., RF, ELM, LightGBM, Cubist, and XGBoost), the pH-hierarchized RF model demonstrated the greatest improvement in predictive accuracy, with a validation R^2 increase of 0.05 after pH-based hierarchization (see Table IV).

The SOM plays a role in adsorbing heavy metals in soil enrichment [82]. After SOM-based hierarchization, the Cubist model exhibited the highest prediction accuracy (validation $R^2 = 0.73$) (see Table IV). This result can be ascribed to the fact that the critical factor affecting the relationship and disparity between soil As and spectral data can be interpreted by SOM information at large scales [83]. The enhanced spectral correlation depicted in Fig. 5, coupled with the conclusive modeling outcomes presented in Table IV, robustly substantiates this distinctive relationship. Some studies have utilized alternative hierarchization methods, including partitioning soil spectra into various subsets according to local elevation, soil type, soil parent material, and soil texture [27], [84]. These methods fully leverage the correlations between soil As and diverse environmental covariates, efficiently coordinating the fluctuations in soil As across different subsets and consequently improving the predictive accuracy. However, the practical application of these hierarchization methods may be significantly limited if obtaining real-time and accurate data for hierarchical variables (such as soil type) proves challenging. Hence, the approach outlined in this study, utilizing precise laboratory SOM measurements, represents a promising solution considering the impact of the SOM on the spectral prediction of soil As.

D. Spatial Distribution of Soil As and Its Influential Factors

The spatial distribution of As concentration in farmland soil in Fig. 13 reveals that high As concentration was mainly concentrated in the central old town area, while low As concentration was primarily located in the northern area surrounding the city center. Long et al. [85] also observed a similar spatial pattern of soil As, indicating that the soil As concentration is often higher near urban central areas. Industrial activities and transportation can affect the heavy metal elements in regional dustfall, further influencing the content of heavy metals in surrounding farmland soil [86]. In Xiangzhou District, urbanization has advanced significantly, with recent years witnessing rapid growth in both agriculture

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TABLE IV
VALIDATION STATISTICS APPLIED TO GLOBAL AND HIERARCHICAL DATASETS DEVELOPED BY MODELING APPROACHES

Strategy	Models	Validation			
		R^2	RMSE	RPIQ	LCCC
Global (non-hierarchication)	RF	0.64	1.19	2.14	0.73
	ELM	0.64	1.18	2.15	0.73
	LightGBM	0.59	1.24	2.08	0.64
	Cubist	0.65	1.17	2.19	0.74
	CNN	0.71	1.02	2.50	0.79
	XGBoost	0.60	1.22	2.08	0.66
Hierarchication-SOM	RF	0.69	1.09	2.34	0.75
	ELM	0.67	1.17	2.17	0.73
	LightGBM	0.60	1.22	2.09	0.68
	Cubist	0.73	0.99	2.57	0.81
	XGBoost	0.64	1.20	2.12	0.71
Hierarchication-pH	RF	0.70	1.08	2.37	0.78
	ELM	0.65	1.23	2.09	0.73
	LightGBM	0.59	1.41	1.81	0.64
	Cubist	0.66	1.18	2.16	0.75
	XGBoost	0.63	1.27	2.01	0.68

and industry, driven by the development of the provincial subcentral city [87]. Industrial systems, primarily those dominated by key sectors such as automobile manufacturing, equipment manufacturing, and economic crop processing, are likely to contribute to a degradation in the nearby farmland soil quality [88]. Xiangzhou District serves as an important transportation hub in Hubei Province, characterized by a high density of road networks and frequent vehicular traffic, which could also have a certain impact on the farmland soil environment near roads due to emissions from vehicle exhaust [19]. In addition, chronic use of agrochemicals and insecticides, combined with the discharge of manufacturing wastewater, could accelerate the accumulation of As in farmland soil within central urban areas.

E. Limitations and Perspectives

In actual agricultural practices, various factors, including climate, terrain, planting systems, and human activities, can all impact the transport of As in farmland soil [63]. Understanding these complex interactions is essential for effective soil management and minimizing contamination risks. Therefore, our future research will focus on exploring the mechanisms that govern the relationships between both natural and anthropogenic factors and the spatial variability of soil As concentration.

In particular, whether the introduction of human activities such as distance to roads, distance to water sources or industrial areas, and the amount of irrigation water applied each year as covariates are expected to further improve the accuracy of soil As concentration spectra prediction deserves further exploration. By analyzing these relationships, we hope to develop more robust predictive models that consider both environmental and anthropogenic influences, ultimately leading to enhanced soil management practices that mitigate contamination risks effectively.

V. CONCLUSION

Using Sentinel-2A images and field-collected soil samples, we investigated the potential of combining spectral hierarchies with different modeling approaches to predict As concentration in farmland soil at a regional scale. Our results demonstrate that the most accurate predictions were achieved by the global CNN model (validated RPIQ = 2.50), followed by the Cubist model (validated RPIQ = 2.19) and the ELM model (validated RPIQ = 2.15). Among the two hierarchical strategies, the SOM-hierarchized Cubist model provided the best predictive performance, achieving a 0.02 improvement in R^2 compared with that of the global CNN model. Notably, deep learning with the CNN model can effectively extract informative local characteristics from soil spectral data, resulting in high-precision predictions of As concentration in farmland soil. SOM-based hierarchization emerged as a prospective strategy for addressing the influence of soil environmental complications on the spectral prediction of As concentration in farmland soil. This is primarily due to the ability of the SOM to effectively capture the spatial pattern of As-spectral relationships influenced by soil properties. In addition, the CMR emerged as the most important predictive variable for As, followed by the B8A and IOR. The identification of high As concentration in the central old town areas underscores the importance of early soil contamination risk warnings on arable lands. In future research, integrating remote sensing images with environmental covariates, such as climate, terrain, and human activities, is anticipated to significantly improve the accuracy of As predictions. This study will contribute to advancing theoretical and technical approaches related to the spectral quantitative prediction of As concentration in farmland soil by leveraging hierarchical modeling strategies.

APPENDIX A
DETAILED DESCRIPTIONS OF THE HYPERPARAMETERS AND THEIR CORRESPONDING OPTIMIZATION RANGES

See Table II.

APPENDIX B
ARCHITECTURE OF THE CNN MODEL

See Fig. 2.

APPENDIX C
EVALUATING METRICS ACROSS MODEL ARCHITECTURES ON VALIDATION DATA

See Figs. 8 and 9.

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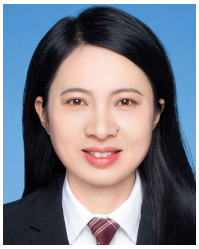
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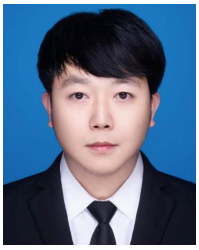
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